

Separable Differential Equations



Solving by separation of variables

- 1 Solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
- 2 Solve $\frac{dy}{dx} = 2xy$, given $y(0) = 5$.

Growth, decay, and applications

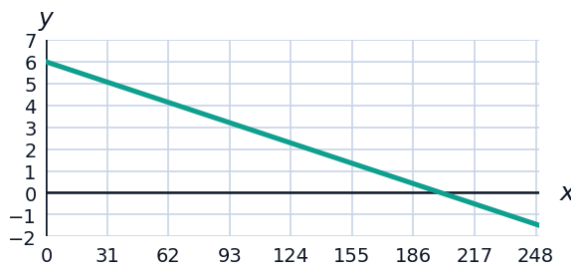
- 3 A population grows according to $\frac{dP}{dt} = 0.05P$, with $P(0) = 2000$.
 - a Solve for $P(t)$.
 - b Find $P(20)$, to the nearest whole number.
- 4 Newton's law of cooling gives $\frac{dT}{dt} = -k(T - 20)$ for an object cooling in a 20°C room. If $T(0) = 100$ and $T(5) = 60$, find k and hence $T(t)$.

More separable equations

- 5 Solve $\frac{dy}{dx} = \frac{y}{x}$, given $y(1) = 4$.
- 6 Solve $\frac{dy}{dx} = y(1 - y)$ (logistic-type equation), given $y(0) = \frac{1}{2}$, using partial fractions $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$.

Mixing and applied models

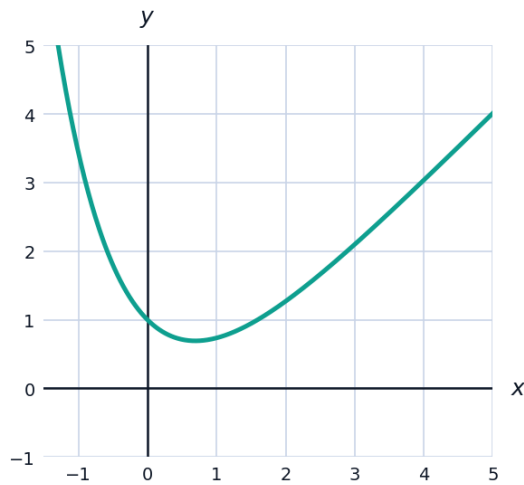
- 7 A tank has 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution drains at the same rate, so the volume stays constant. Let $y(t)$ be the amount of salt (kg) at time t . The differential equation is $\frac{dy}{dt} = 6 - \frac{3y}{100}$.



- a Find the equilibrium value of y (where $\frac{dy}{dt} = 0$).
- b Explain, in words, what this equilibrium represents physically.

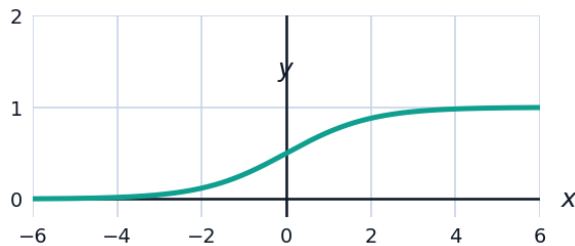
Slope fields and qualitative behavior

- 8 For the differential equation $\frac{dy}{dx} = x - y$, find the slope of the solution curve at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$.
- 9 Verify that $y = x - 1 + 2e^{-x}$ is a solution to $\frac{dy}{dx} = x - y$, by substituting into both sides.



Visualizing a logistic solution curve

- 10 The graph shows the logistic solution $y = \frac{e^x}{1 + e^x}$ found earlier for $\frac{dy}{dx} = y(1 - y)$, $y(0) = \frac{1}{2}$. Use the graph to describe the long-run behavior of y as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.



Another applied separable equation

- 11 Radioactive decay follows $\frac{dA}{dt} = -kA$. If a sample decays from 80 g to 20 g in 30 days, find k and the half-life of the substance.

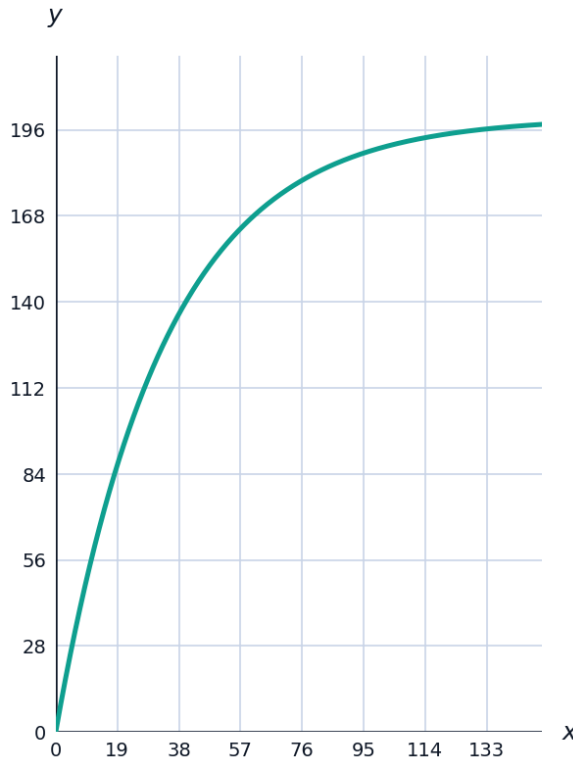
Separable equations with algebraic solutions

12 Solve $\frac{dy}{dx} = \frac{x^2}{y^2}$, given $y(0) = 2$.

13 Solve $\frac{dy}{dx} = \frac{1}{2y}$, given $y(0) = 3$.

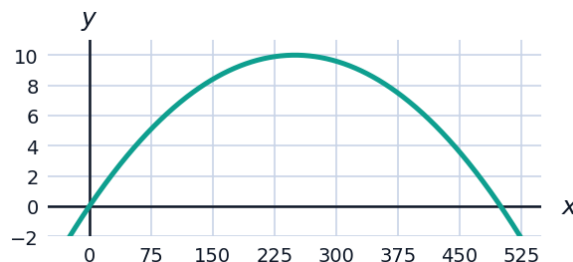
Verifying a solution to an applied model

- 14 For the mixing problem $\frac{dy}{dt} = 6 - \frac{3y}{100}$ with $y(0) = 0$, verify that $y(t) = 200(1 - e^{-0.03t})$ is a solution, and confirm it approaches the equilibrium 200 found earlier as $t \rightarrow \infty$.



15 Solve $\frac{dy}{dx} = e^{x-y}$, given $y(0) = 0$, using the fact that $e^{x-y} = e^x \cdot e^{-y}$.

- 16 A pond's fish population P grows according to $\frac{dP}{dt} = 0.08P(1 - \frac{P}{500})$, a logistic model with carrying capacity 500. Without solving the equation explicitly, determine the equilibrium population levels (where $\frac{dP}{dt} = 0$), and state which one is realistic for a growing population starting from $P(0) = 50$.



17 Solve $\frac{dy}{dx} = \frac{\cos x}{y}$, given $y(0) = -2$, and state why the negative root must be chosen.

A final applied separable model

- 18** A drug is eliminated from the bloodstream according to $\frac{dC}{dt} = -0.2C$, where C is concentration (mg/L) and t is hours. If the initial dose gives $C(0) = 50$ mg/L, find $C(t)$, then find how many hours it takes for the concentration to drop below 5 mg/L, to one decimal place.